

Functions assignment

Q1. What is the difference between a function and a method in Python?

Ans. a function is not always present/ not defined in the class but the method is the function can be called by name and the method will be called by object The function operates on the data that you pass to them as an argument and method It operates on the data of the object it associates with

Q2. Explain the concept of function arguments and parameters in Python.

Ans. Function parameters are the names listed in the function's definition. Function arguments are the real values passed to the function The function has been but through parenthesis and the argument are specified after function

Q3. What are the different ways to define and call a function in Python.

Ans. to define function we use keyword "def" Such as `def functiona(x):` use the def keyword, name your function, add optional parameters in parentheses, write your code, and optionally return a value.

Q4. What is the purpose of the `return` statement in a Python function.

Ans. return is used to end the execution of the function call and it "returns" the value of the expression following the return keyword to the caller. The statements after the return statements are not executed. If the return statement is without any expression, then the special value None is returned

Q5. What are iterators in Python and how do they differ from iterables.

Ans. explaining with an example. # If you want to cook food so you will cook food which will be cooked #so something that can be cooked that is known as iterator #the process of cooking that is iteration #the vegetable is iterable #so that stage that can be cooked that is known as iteration

Q6. Explain the concept of generators in Python and how they are defined.

Ans allows you to declare a function that behaves like an iterator, providing a faster and easier way to create iterators and a routine that can be used to control the iteration behaviour of a loop.

Q7. What are the advantages of using generators over regular functions.

Ans. memory efficiency, readability, and speed. And uses less memory space.

Q8. What is a lambda function in Python and when is it typically used.

Ans LMBDA is "Language Model for Dialogue Applications". The lambda function performs both addition and multiplication and returns a tuple with both results. This is useful for scenarios were multiple.

Q9. Explain the purpose and usage of the `map()` function in Python.

Ans. The `map()` function in Python applies a given function to each item in an iterable (like a list, tuple, or set) and returns a map object, which is an iterator. This allows you to transform or process all items in an iterable without needing an explicit for loop.

Q10. What is the difference between `map()`, `reduce()`, and `filter()` functions in Python?

Ans. The `map()` function iterates through all items in the given iterable and executes the function we passed as an argument on each of them. The `filter()` forms a new list that contains only elements that satisfy a certain condition, i.e. the function we passed returns True. The `reduce()` works differently than `map()` and `filter()`. It does not return a new list based on the function and iterable we've passed. Instead, it returns a single value. Also, in Python 3 `reduce()` isn't a built-in function anymore, and it can be found in the `functools` module.